

NAKHON SI THAMMARAT, Thailand

WMO: 485520

Lat: 8.5461N

Lon: 99.9396E

Elev: 6

StdP: 101.26

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	21.5	22.3	19.5	14.3	25.9	20.3	15.0	26.1	6.1	27.5	5.3	27.8	0.1	330	0.299

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	8.3	35.0	26.3	34.3	26.2	33.7	26.2	27.6	32.9	27.2	32.3	27.0	31.9	2.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.2	21.6	29.6	25.9	21.3	29.4	25.7	20.9	29.1	87.5	33.1	86.1	32.6	84.8	32.1	32.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
5.1	4.3	3.7	DB	20.1	36.1	1.1	0.7	19.3	36.6	18.6	37.0	18.0	37.4	17.2	37.9	
			WB	18.7	28.6	1.5	1.2	17.6	29.4	16.7	30.1	15.9	30.8	14.8	31.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.9	26.4	26.9	27.8	28.9	29.2	29.0	28.8	28.7	28.3	27.6	26.8	26.4
	DBStd	1.42	0.92	0.94	1.01	1.10	0.97	0.86	0.89	0.92	0.83	1.01	1.17	1.00
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6533	508	472	553	567	596	571	583	578	549	545	503	507
	CDD18.3	3492	250	239	295	317	338	321	324	320	299	287	253	249
	CDH23.3	35181	2137	2300	3145	3648	3768	3574	3494	3371	2946	2637	2145	2016
	CDH26.7	13464	608	833	1304	1641	1634	1526	1453	1379	1110	872	598	506
Wind	WSAvg	0.8	0.9	1.0	1.0	0.8	0.7	0.7	0.7	0.7	0.6	0.6	0.7	0.9
Precipitation	PrecAvg	2520	236	65	52	86	180	83	109	90	165	367	606	480
	PrecMax	3593	1240	179	118	229	317	198	177	170	330	640	875	871
	PrecMin	1676	14	2	0	17	30	34	19	24	61	100	332	195
	PrecStd	482	302	55	38	70	80	45	51	44	78	137	171	212
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.9	33.2	34.5	35.9	35.9	35.2	35.1	35.0	34.2	33.2	32.2	31.6
		MCWB	25.6	25.9	26.4	27.2	27.0	25.9	25.6	25.4	25.8	26.3	26.1	25.7
	2%	DB	31.0	32.1	33.4	34.7	34.7	34.3	34.1	34.0	33.2	32.3	31.4	30.8
		MCWB	25.2	25.4	26.1	26.9	26.8	26.1	25.7	25.6	25.9	26.1	26.0	25.5
	5%	DB	30.3	31.3	32.6	33.8	33.9	33.6	33.3	33.2	32.5	31.7	30.8	30.2
		MCWB	25.0	25.1	25.9	26.6	26.7	26.2	25.8	25.7	25.9	26.0	25.9	25.3
	10%	DB	29.7	30.7	31.9	33.0	33.0	32.8	32.5	32.4	31.7	30.9	30.0	29.4
		MCWB	24.8	24.9	25.7	26.4	26.6	26.2	25.7	25.7	25.8	25.9	25.7	25.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.6	26.5	27.2	28.1	28.0	27.7	27.2	27.2	27.2	27.2	27.2	26.9
		MCDB	30.3	31.7	32.6	34.4	33.7	32.9	32.2	32.2	31.9	31.4	30.7	30.1
	2%	WB	26.0	26.0	26.7	27.4	27.5	27.1	26.7	26.7	26.7	26.7	26.7	26.3
		MCDB	29.6	30.8	31.9	33.2	33.0	32.3	31.7	31.7	31.3	30.7	30.0	29.4
	5%	WB	25.6	25.6	26.3	27.1	27.2	26.8	26.4	26.4	26.4	26.4	26.3	25.9
		MCDB	29.0	30.1	31.3	32.4	32.4	31.8	31.3	31.3	30.9	30.3	29.5	29.0
	10%	WB	25.2	25.3	26.0	26.7	26.8	26.4	26.1	26.0	26.0	26.1	26.0	25.5
		MCDB	28.4	29.5	30.8	31.7	31.7	31.3	30.9	30.9	30.5	29.9	29.1	28.2
Mean Daily Temperature Range	5% DB	MDBR	6.6	8.1	8.5	8.6	8.3	8.2	8.1	8.2	7.8	6.8	5.7	5.8
		MCDBR	7.7	9.0	9.6	9.8	9.4	9.3	9.2	9.3	8.8	8.1	7.4	7.3
		MCWBR	2.9	3.3	3.3	3.2	2.8	2.7	2.6	2.7	2.7	2.8	2.8	2.8
	5% WB	MCDBR	6.8	8.1	8.8	9.1	8.8	8.4	8.4	8.6	8.2	7.4	6.6	6.3
		MCWBR	2.9	3.2	3.3	3.2	3.0	2.8	2.8	2.9	2.9	2.8	2.8	2.7
Clear-Sky Solar Irradiance	taub	0.494	0.500	0.507	0.478	0.454	0.452	0.453	0.447	0.439	0.453	0.476	0.486	
	taud	2.167	2.144	2.160	2.248	2.354	2.391	2.373	2.380	2.392	2.355	2.270	2.211	
	Ebn at Noon	821	830	827	839	842	834	836	855	872	860	832	819	
	Edh at Noon	151	159	158	143	125	119	122	124	123	127	135	142	
All-Sky Solar Radiation	RadAvg	4.56	5.54	5.75	5.73	5.12	4.92	4.84	4.98	4.88	4.37	3.91	3.89	
	RadStd	0.44	0.38	0.50	0.35	0.29	0.26	0.29	0.25	0.19	0.36	0.38	0.40	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.16	N/A	N/A	N/A	+0.23	N/A	N/A	N/A	N/A
(47)	Regional (16 neighbors)	+0.20	+0.27	N/A	+0.28	+0.21	N/A	N/A	+74	+75

Nomenclature: See separate page